

ENEL700 Tutorial 8

1. Can we use a transmitter as a receiver? Justify your answers with simple sketches.
2. An amplifier operating over the frequency range of 455 to 460kHz has a $200\text{k}\Omega$ input resistor. What is the RMS noise voltage at the input to this amplifier if the ambient temperature is 17°C ?
3. A receiver with a 75Ω input resistance operates at a temperature of 31°C . The received signal is at 89MHz with a bandwidth of 6MHz. The received signal voltage of $8.3\mu\text{V}$ is applied to an amplifier with a noise figure of 2.8dB. Find the output signal-to-noise ratio in dB.
4. Calculate the noise figure (NF) of a cascade connections of amplifier and a mixer the amplifier has a $\text{NF}=5\text{dB}$ and a gain of 20dB, while the mixer stage has a $\text{NF}=12\text{dB}$ and a gain of -2dB.